McGill

Where to Begin

Explainable AI (XAI)

Presented by

Tamires Marcal

linkedin.com/in/tamiresco

tamires.marcal@mail.mcgill.ca

McGill

Explainable AI (XAI): Where to Begin

Today's Journey

- Why Explainability Matters
- Examples
- Toolkit
- Deepdive: SHAP Values
- How to Troubleshoot
- Key Takeaways

linkedin.com/in/tamiresco
tamires.marcal@mail.mcgill.ca

Why Explainability Matters

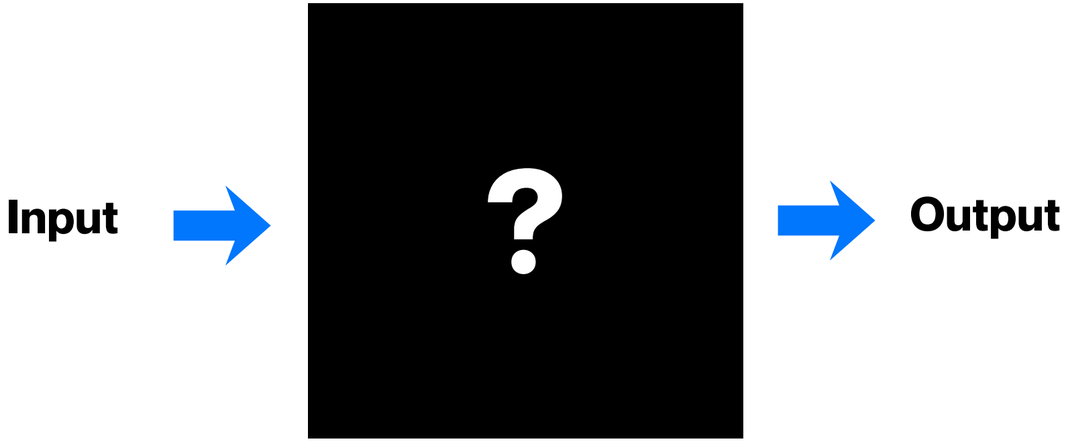
McGill

Traditional AI models are opaque

- Input goes in
- Decision comes out
- No understanding of WHY

This creates serious challenges when

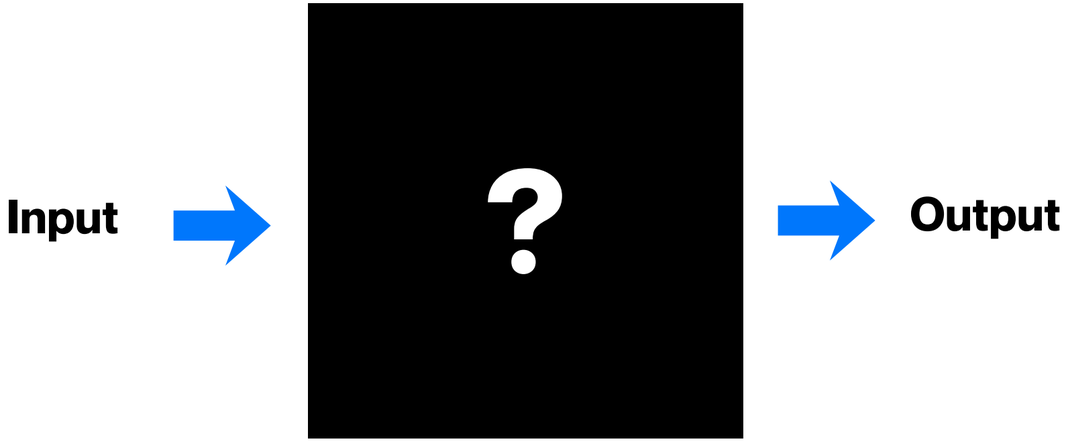
- Decisions affect people's lives
- Compliance is required
- Trust is essential



linkedin.com/in/tamiresco
tamires.marcal@mail.mcgill.ca

Transparency is important for several roles

- consumers want to know how AI makes decisions
- executives say AI transparency is critical
- some regulated industries require explainability



linkedin.com/in/tamiresco
tamires.marcal@mail.mcgill.ca

McGill

Key Concepts

Blackbox
Model whose internal decision process is not understandable to human

Transparency
Degree to which the model's structure, data, and processes are accessible

Interpretability
Ability to understand the model's internal mechanisms fully

Explainability
Ability to provide human-understandable reasons for specific predictions

Fairness
Ensuring model decisions do not produce systematic bias

linkedin.com/in/tamiresco
tamires.marcal@mail.mcgill.ca

McGill

What can you do with Explainability

Developing & Troubleshooting

- Data Leaking
- Bias Detection
- Spurious Correlations
- Performance Drops

Discovering & Learning

- Specialist knowledge gain → Insights
- Only if performance is going well

Human Trust & Adoption

- Build a Model that people trust
- Build a Narrative

Legal & Regulatory

- Mandatory in some cases
- Preparing for Edge Cases

linkedin.com/in/tamiresco
tamires.marcal@mail.mcgill.ca

Examples

McGill

Financial Risk Model

Use Case: Credit Scoring & Loan Approval

Challenge

Banks must explain why a loan application was denied

XAI Solution

Low credit score contributed -35 points

High debt-to-income ratio contributed -22 points

Recent bankruptcies contributed -18 points

Outcome

Regulatory compliance + Actionable feedback for applicants

linkedin.com/in/tamiresco

tamires.marcal@mail.mcgill.ca

McGill

Retail Demand Forecast

Use Case: Estimating the Quantity of Items to be Sold

Challenge

Retail companies seek to estimate sales and understand their drivers

XAI Solution

Increase in outdoor gear sales (40% influence)
Decrease in Temperature (30% influence)
Seasonal trends favor camping equipment (20% influence)
Reduce price (20% influence)

Outcome

Inventory optimization
Marketing Insights
Pricing Strategies
Seasonal Influence in Finance

linkedin.com/in/tamiresco
tamires.marcal@mail.mcgill.ca

McGill

Medical Diagnosis

Use Case: Cancer Detection from Imaging

Challenge

Understanding the model so that the hospital, regulatory agencies and physicians start to trust

XAI Solution

Heat map shows suspicious tissue areas
Identifies specific features the model focused on
Doctors can verify AI reasoning against medical knowledge

Outcome

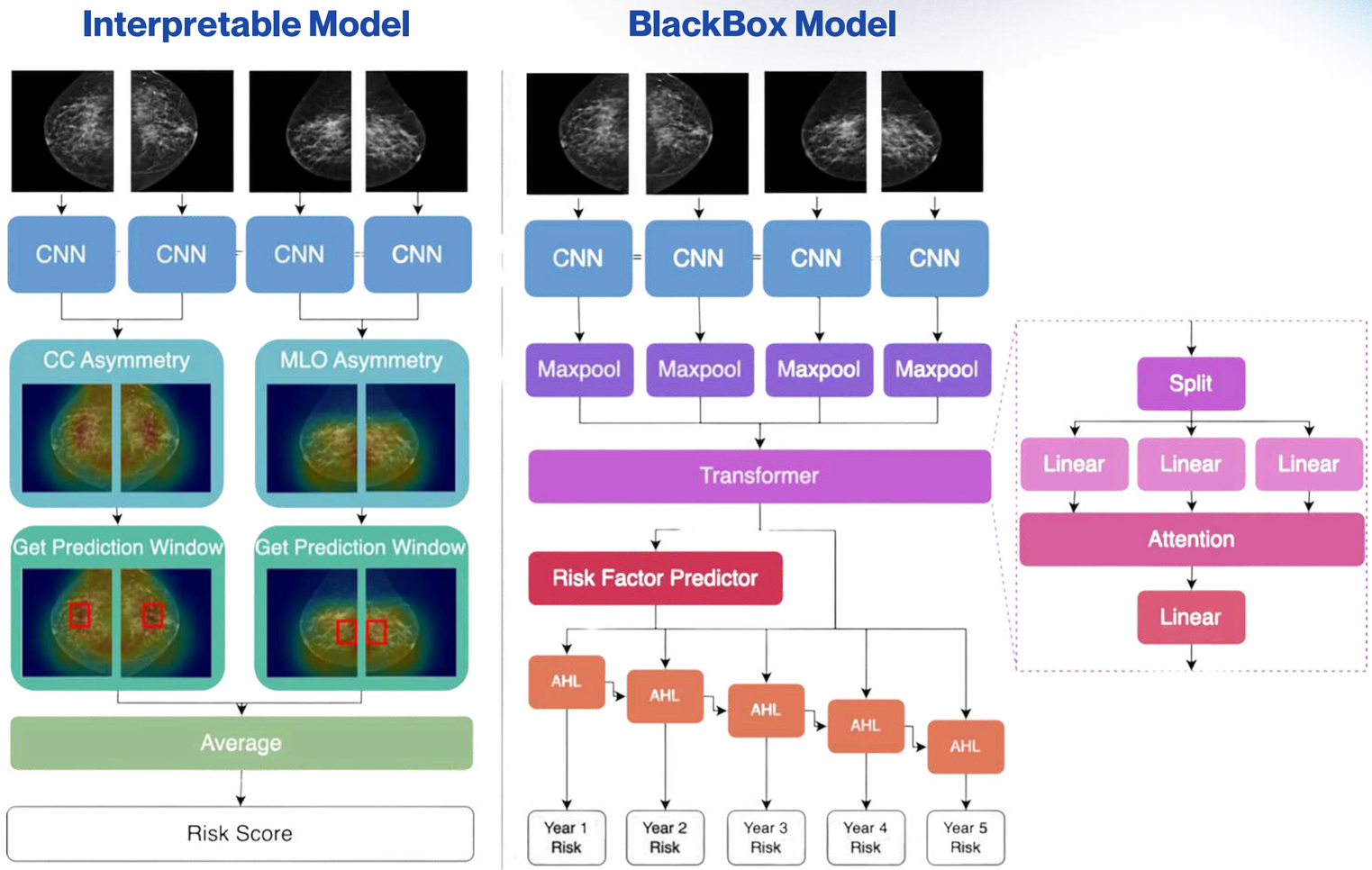
Increased physician confidence + Better patient care

linkedin.com/in/tamiresco
tamires.marcal@mail.mcgill.ca

McGill

Medical Diagnosis

Use Case: Cancer Detection from Imaging



linkedin.com/in/tamiresco

tamires.marcal@mail.mcgill.ca

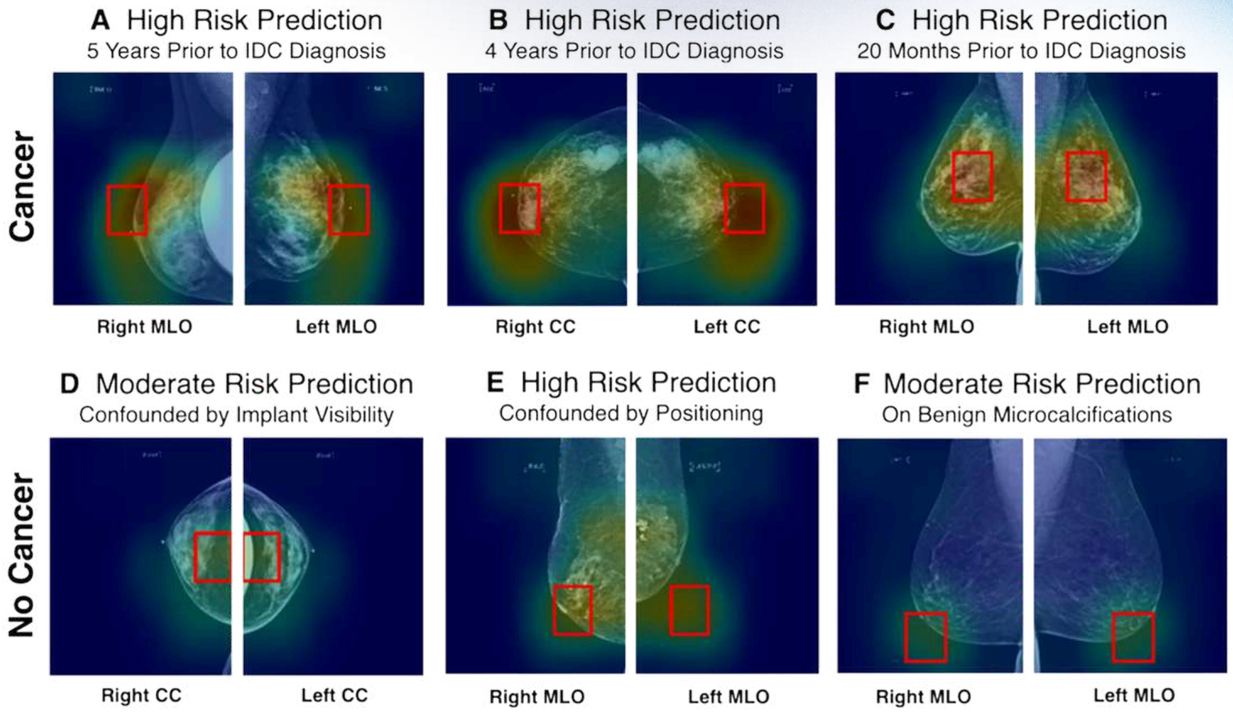
McGill

Medical Diagnosis

Use Case: Cancer Detection from Imaging

AsymMirai: Interpretable Mammography-based Deep Learning Model for 1-5-year Breast Cancer Risk Prediction

Jon Donnelly [1], Luke Moffett [1], Alina Jade Barnett [1], Hari Trivedi [1], Fides Schwartz [1], Joseph Lo [# 1], Cynthia Rudin [# 1]



	1-year AUC	2-year AUC	3-year AUC	4-year AUC	5-year AUC
Interpretable Model	0.79 (0.73, 0.85)	0.69 (0.65, 0.73)	0.68 (0.65, 0.71)	0.67 (0.64, 0.70)	0.66 (0.63, 0.69)
BlackBox Model	0.84 (0.79, 0.89)	0.74 (0.70, 0.78)	0.72 (0.69, 0.76)	0.72 (0.69, 0.75)	0.71 (0.68, 0.74)

linkedin.com/in/tamiresco
tamires.marcal@mail.mcgill.ca

McGill

Image Classifier

Use Case: Troubleshooting

Challenge

Why is there poor performance in classifying horses?

XAI Solution

Heatmap

Debugging

Outcome

Train again with changes in the dataset

linkedin.com/in/tamiresco

tamires.marcal@mail.mcgill.ca

McGill

Image Classifier

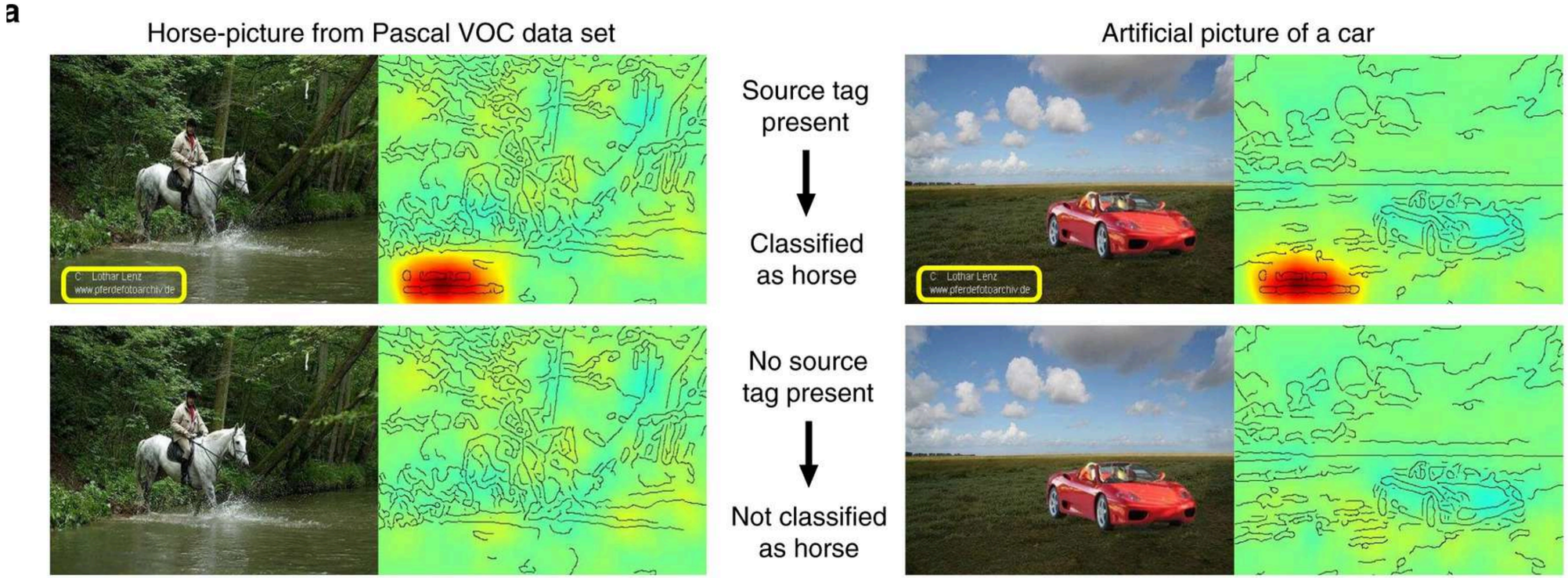
Use Case: Troubleshooting

Article | Open access | Published: 11 March 2019

Unmasking Clever Hans predictors and assessing what machines really learn

Sebastian Lapuschkin, Stephan Wäldchen, Alexander Binder, Grégoire Montavon, Wojciech Samek & Klaus-Robert Müller

Nature Communications **10**, Article number: 1096 (2019) | Cite this article



linkedin.com/in/tamiresco
tamires.marcal@mail.mcgill.ca

Toolkit

McGill

Model-Specific

Designed for specific models

1. Linear Models (coefficient inspection)
2. Decision Trees (inherently interpretable)
3. Grad-CAM (neural networks)
4. Attention weights (transformers)

Model Agnostic

Work with ANY model

1. SHAP (Shapley values)
2. LIME (Local interpretability)
3. Partial Dependence Plots
4. Permutation Importance

linkedin.com/in/tamiresco
tamires.marcal@mail.mcgill.ca

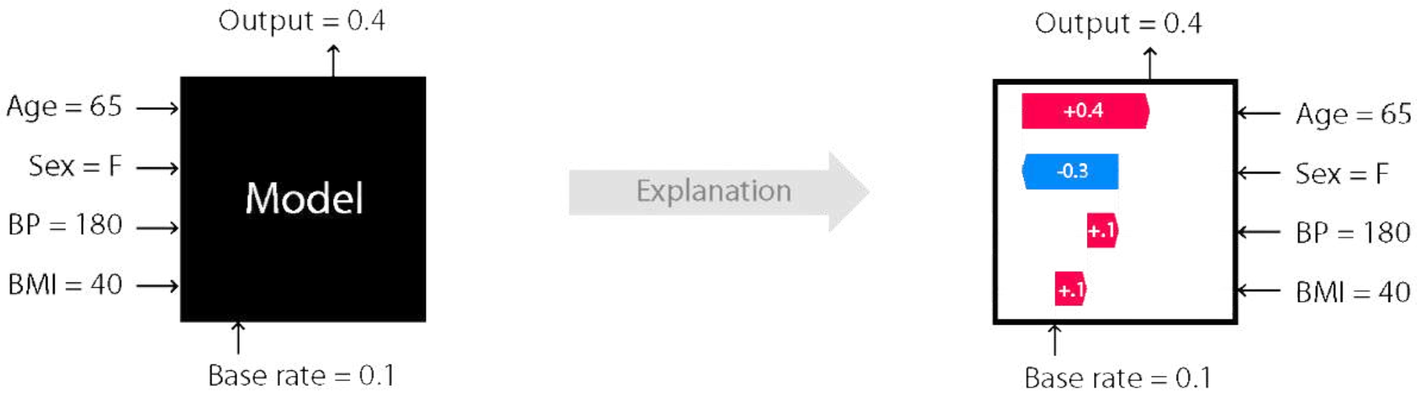
Deepdive: SHAP Values

McGill

Deepdive: SHAP Values

SHAP (SHapley Additive exPlanations)

Based on Shapley values from cooperative game theory



Contributions are:

- Additive
- Consistent
- Fair (theoretical guarantee)

Why SHAP?

- Works for any model (tree, neural network, linear)
- Local explanations → can aggregate to global insights
- Widely used in healthcare, finance, and scientific ML

linkedin.com/in/tamiresco

tamires.marcal@mail.mcgill.ca

McGill

Deepdive: SHAP Values

1. **Install SHAP**

```
pip install shap
```

2. **Create Explainer**

```
explainer = shap.TreeExplainer(model)
```

3. **Generate Values**

```
shap_values = explainer.shap_values(X)
```

4. **Visualize Results**

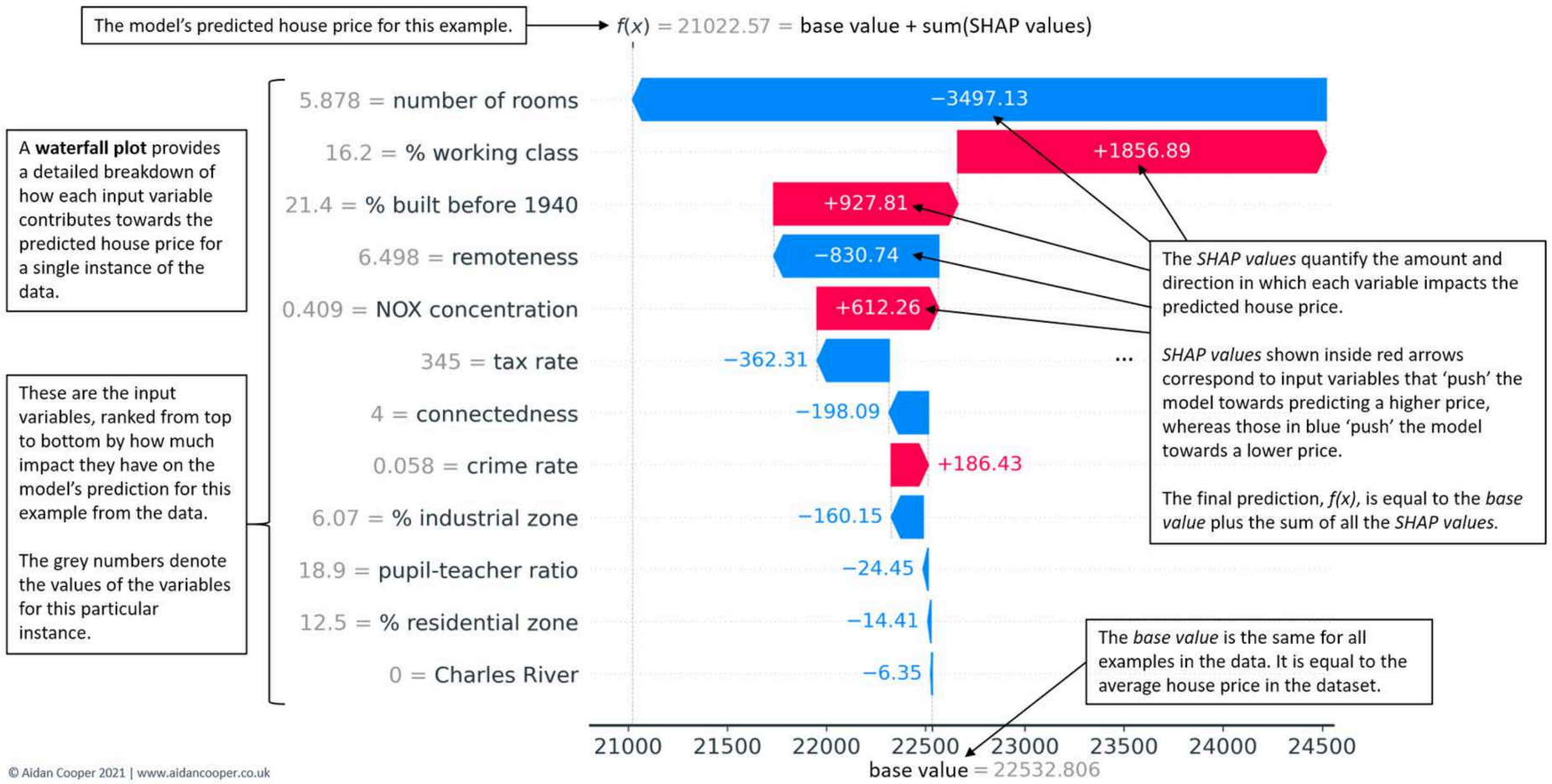
```
shap.summary_plot(shap_values, X)
```

linkedin.com/in/tamiresco
tamires.marcal@mail.mcgill.ca

McGill

Deepdive: SHAP Values

Waterfall Plot

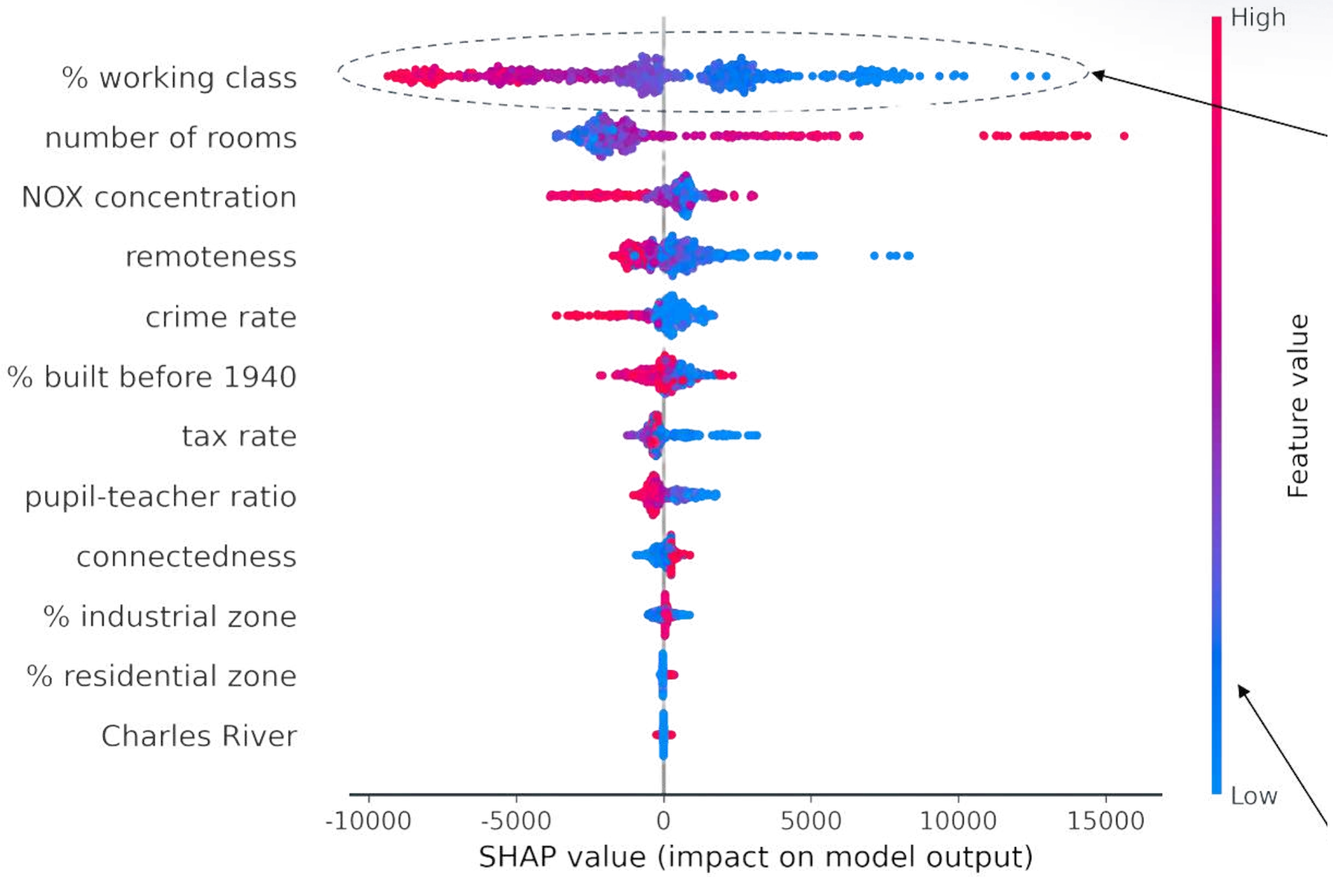


n/tamiresco
tamires.marcal@mail.mcgill.ca

McGill

Deepdive: SHAP Values

Beeswarm Plot



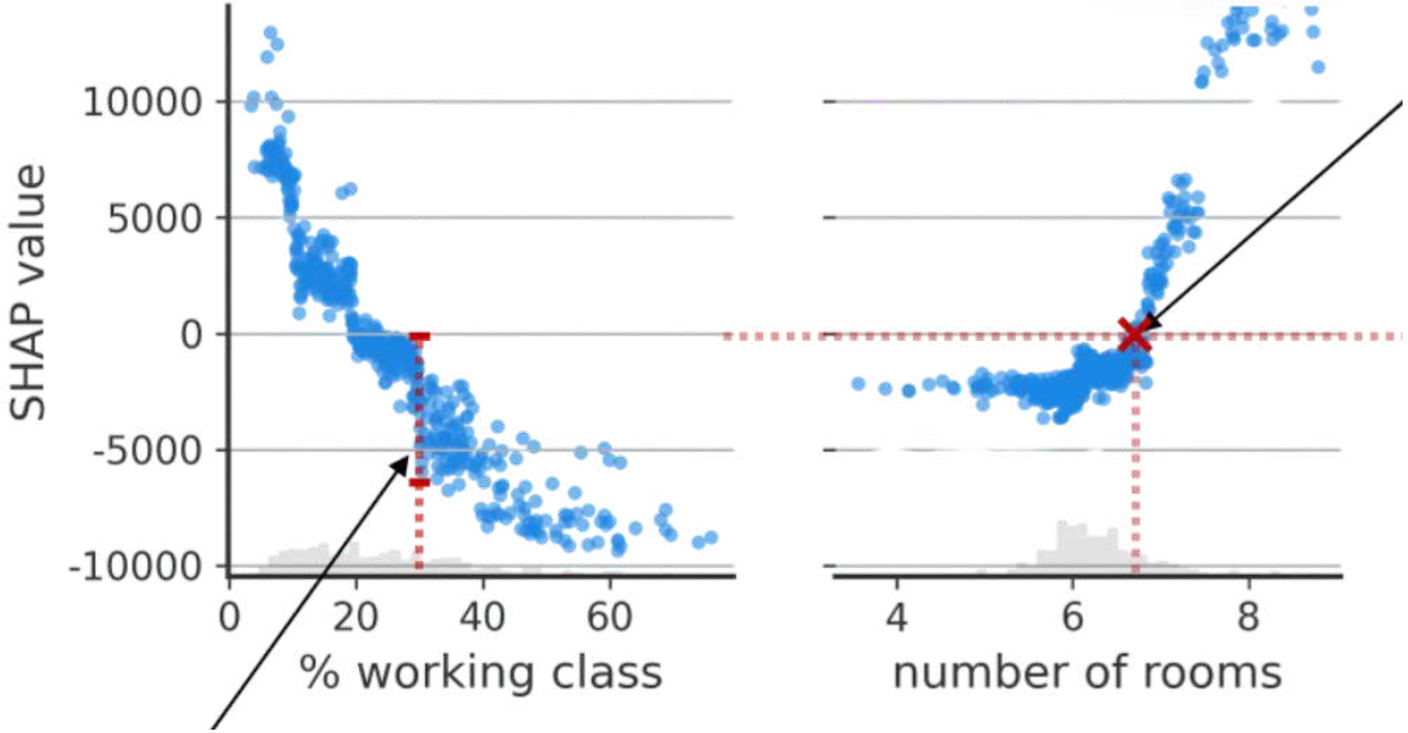
© Aidan Cooper 2021 | www.aidancooper.co.uk

linkedin.com/in/tamiresco
tamires.marcal@mail.mcgill.ca

McGill

Deepdive: SHAP Values

Dependence Plot

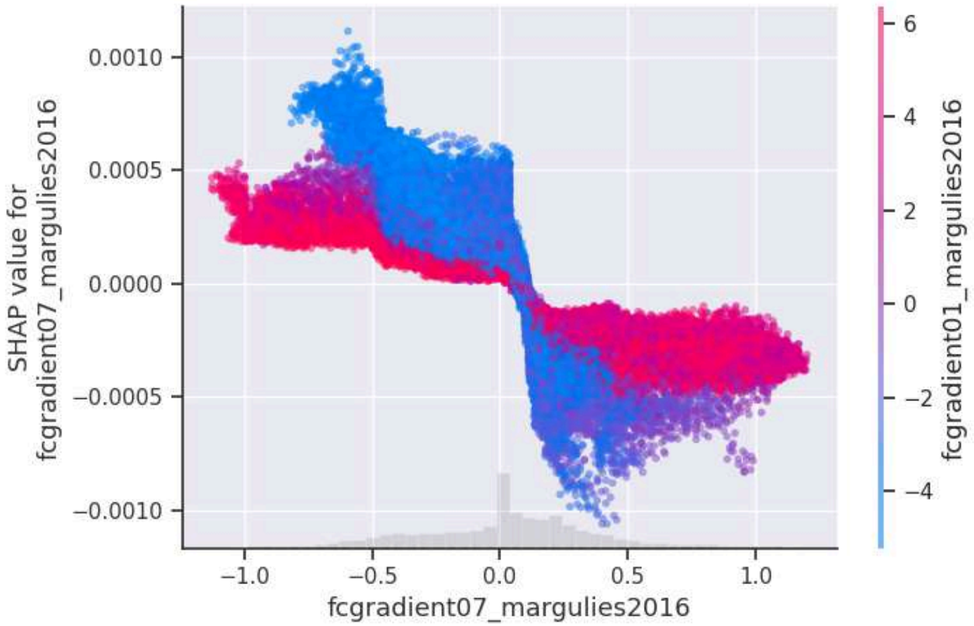


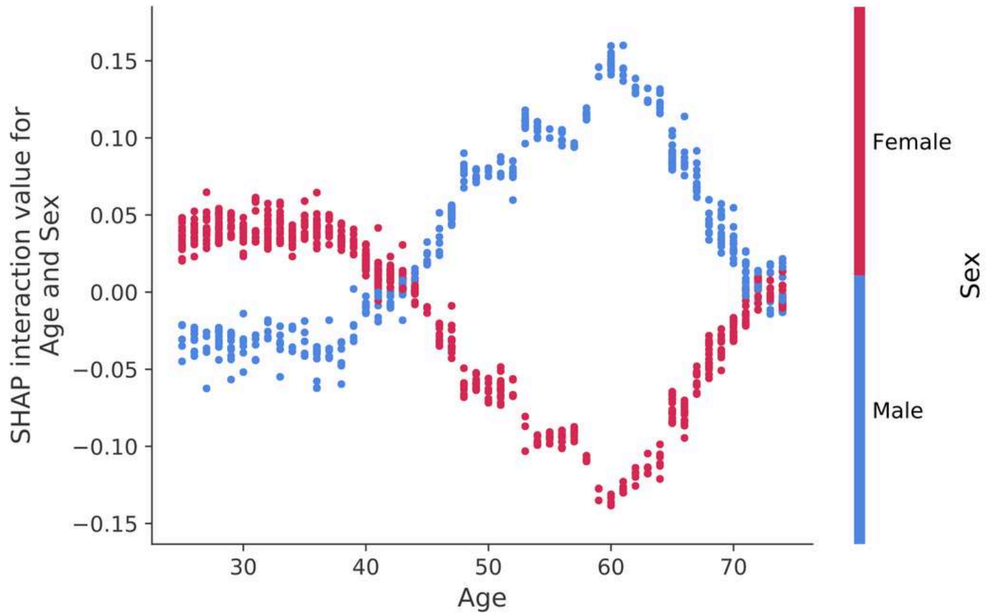
linkedin.com/in/tamiresco
tamires.marcal@mail.mcgill.ca

McGill

Deepdive: SHAP Values

Interactions



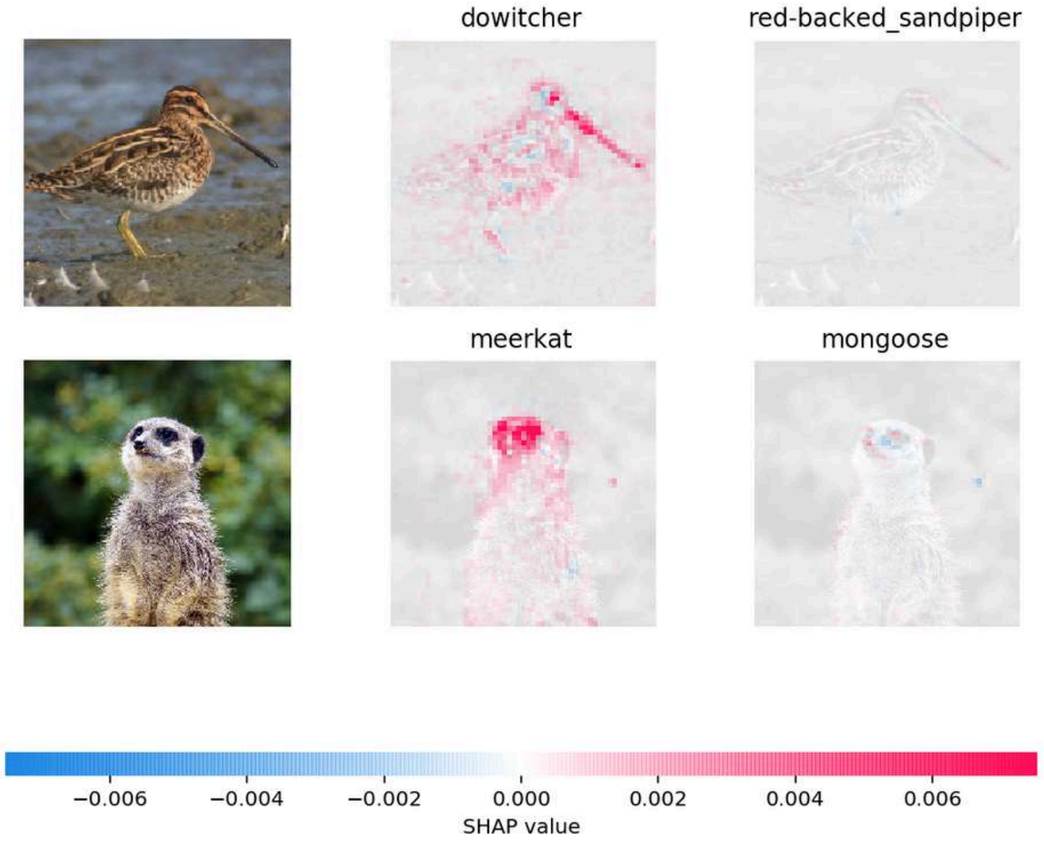


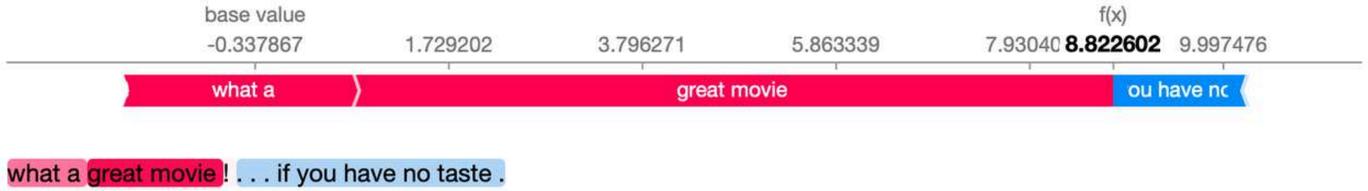
linkedin.com/in/tamiresco
tamires.marcal@mail.mcgill.ca

McGill

Deepdive: SHAP Values

Images & Text





linkedin.com/in/tamiresco
tamires.marcal@mail.mcgill.ca

How to Troubleshoot

McGill

Common Problems

When performance changes or behavior looks suspicious

1) Data Leakage

- A feature dominates predictions with unrealistically high impact
- Example: ID fields, timestamps, post-outcome variables
- Signal: One feature with extreme SHAP values across most samples

2) Bias & Fairness Issues

Sensitive or proxy variables drive predictions (e.g., ZIP code, income, gender proxies)
Signal: Large SHAP contributions correlated with protected groups
Helps audit group-level impact differences

3) Spurious Correlations

Model relies on artifacts instead of meaningful signals
(e.g., background patterns, hospital ID, device type)
Signal: High importance for features that shouldn't be causally relevant
Useful for detecting "Clever Hans" behavior

4) Performance Drops / Model Drift

Feature importance changes between training and production
Signal: SHAP distribution shift over time
Helps identify data drift or concept drift

5) Model Overfitting

Model relies heavily on rare or noisy features
Signal: Large SHAP values for low-frequency variables
Often appears as unstable explanations across samples

6) Unexpected Feature Effects

Nonlinear or counterintuitive relationships
Example: higher risk predicted for healthier patients
Signal: SHAP dependence plots showing reversed or irregular patterns

linkedin.com/in/tamiresco
tamires.marcal@mail.mcgill.ca

McGill

XAI-Powered Debugging Workflow

1. Train your model as usual
2. Apply XAI method (SHAP, LIME, etc.)
3. Analyze feature importance & predictions
4. Identify suspicious patterns or features
5. Investigate data quality or feature engineering
6. Retrain and validate improvements

linkedin.com/in/tamiresco
tamires.marcal@mail.mcgill.ca

Key Takeaways

McGill

Key Takeways

Developing & Troubleshooting

- Data Leaking
- Bias Detection
- Spurious Correlations
- Performance Drops

Discovering & Learning

- Only if performance is going well
- Specialist knowledge gain → Insights

Human Trust & Adoption

- Build a Model that people trust
- Build a Narrative

Legal & Regulatory

- Mandatory in some cases
- Preparing for Edge Cases

linkedin.com/in/tamiresco
tamires.marcal@mail.mcgill.ca

McGill

Thank you!

McGill

neuro

Mila

MCIN MCGILL CENTRE for INTEGRATIVE NEUROSCIENCE

HEALTHY BRAINS HEALTHY LIVES

HIBALL HELMHOLTZ International BigBrain Analytics & Learning Laboratory

linkedin.com/in/tamiresco

tamires.marcal@mail.mcgill.ca